Test Method for Laminar Peel Strength	Test Method #: VF- F04 REF # :TM-R-029 Page 1 of 4
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PURPOSE AND SCOPE: To measure the level of adhesion between two materials which have been bonded together, test samples are prepared using controlled conditions and techniques and are evaluated using the 180º peel strength (T-peel) technique.

REFERENCES: DIN 53273 SATRA AM 1

EQUIPMENT AND MATERIAL:

1. Tensile testing machine, meeting the requirements of ASTM D638
2. Clamping jaws for the tensile testing machine, which will not damage the materials being tested.
3. Steel scale, capable of measuring to 0.5mm.
4. Other required equipment will vary dependent on the types of material being bonded and tested
5. Press capable of applying the specified pressure.
6. Lining materials to upper materials, foam to upper materials, fabric to pump bladders, upper to non-sew component, upper to molded or injection logo, sock-liners, etc.

SAMPLE PREPARATION:

Dependent on the combination of materials and adhesives being combined, a wide variety of sample preparation techniques may be used .If a sample preparation technique for the materials being evaluated is not described below, the samples should be prepared according to the directions of the material supplier or as they would be used in a shoe factory .The bonding procedure used for such materials should be described in the test report.

A) MATERIALS COATED WITH A PRESSURE –SENITIVE ADHESIVE:

1. Position one substrate material onto the other substrate material such that the two materials are wrinkle-free.
2. Press the two materials together for 10-15 seconds with a pressure of 1-2 kgf/cm2, (one to two kgf load for every square centimeter of surface area).
3. Condition the samples for 24 hours at 23+/-2ºC and 55+/-10% relative humidity.
4. Following conditioning, die or hand cut test samples to width of 25+/-1mm.
5. Initiate at least 2 cm between the 2 substrate materials for the clamp to grip.

B) SOLVENT –ACTIVATED HEEL COUNTER:

1. Die cut a 2.5+/-0.1cm wide strip from the heel counter. The length of the strip should be approximately 10cm.
   Note: With pre-molded /injection –molded heel counters this should be done in the horizontal direction, and close to the base of the heel counter

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2. Obtain a piece of 1.3+/-0.2mm full grain leather which is a minimum of 3.5cm wide by 12 cm long.
3. Dip the heel counter strip in acetone for two (2) seconds.
4. Immediately upon removal from the acetone, place the heel counter strip onto the "flesh side" of the leather and press between two flat metal plates which are a minimum of 5cm in width and 15cm length. A load of 25-50 kgf should be placed on top of the samples for 30 to 60 minutes.
5. Condition the samples for 24 hours at 23+/-22ºC and 55+/-10% relative humidity.
6. Following conditioning, cut the excess width of leather from the test sample using a razor blade (Final width of the heel counter/leather test sample should be 2.5+/-0.1cm).
7. Initiate at least 2 cm between the heel counter and leather or any substrate material for the clamp to grip.

C. HEAT-ACTIVATED HEEL COUNTER OR BOX TOE:

1. Die cut a 2.5+/-0.1cm wide by 15cm long strip from the heel counter.
2. Obtain a piece of 1.3+/-0.2mm full grain leather which is a minimum of 3.5cm wide by 20cm long.
3. Position the heel counter test sample onto the "flesh side" of the leather test sample.
4. Press the two materials together using a pressure of 2.0+/-0.5 kgf /cm2, a temperature of 180 ºC, and a dwell time of 12-15 seconds.
5. Condition the samples for 24 hours at 23+/-22ºC and 55+/-10% relative humidity.
6. Following conditioning, cut the excess width of leather from the test sample using a razor plate, (Final width of the test sample should be 2.5+/-0.1cm).
7. Initiate at least 2 cm between the heel counter or box toe and leather or any substrate material for the clamp to grip.

D. Sock-liners :

Option 1:

1. Die-cut samples which are 2.5+/-0.1cm in width by approximately 15cm in length.
2. Initiate at least 2 cm between the fabric cover from the foam for the clamp to grip.

Option 2:

1. Test the laminar peel strength of the whole piece of sock-liner – don't cut into strip test piece.
2. Initiate at least 2 cm between the fabric cover from the foam for the clamp to grip.

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E. Upper with Non-Sew Component or Any Material with Injection or Molded Component

1. Test the laminar peel strength of the component which is bonded to the upper / material. The upper or material can be cut into smaller piece as long as the non-sew or logo or the small component bonded to upper or material not affected.
2. Initiate cut in between the bonded components for the clamp to grip.

REPETITIONS: Prepare three (3) bond samples per adhesive or material combination.

TEST PROCEDURE:

Test should be conducted at standard laboratory conditions @ 23.0±2.0˚C temperature and 55 ±10 % Relative Humidity.

A) Dry Test

1. To have accurate spacing during testing, put a mark on the testing sample approximately every 2cm and number these locations starting at the initiation point in ascending order.
2. Measure the width of the test sample to the nearest 0.05cm at every 2 cm marked. This will be used to calculate force per unit width.
3. Zero and calibrate the tensile testing machine. Set the crosshead speed to 100mm/minute, the jaw gap to approximately 3 cm, and measure, the load range can be increased to 20kgf).
4. Centrally clamp approximately 2 cm of the free end of the “stretcher” substrate of the bond sample into the stationary jaw of the tensile tester, centrally clamp approximately 2 cm of the free end of the second substrate of the bond sample into the moving jaw of the tensile tester. The sample must be inserted into the jaws such that the bond line can be observed by the operator as peeling progresses.
5. Operate the tensile testing machine at a speed of 100mm/minute.
6. Continue testing until the bond sample has been completely separated.
7. Calculate the force required to pull the test specimen apart **at every 2 cm marked** by dividing the force by the measured width of the test specimen. If an automated test system is used, this can be done automatically. For manual recording systems, the bonding strength should be calculated from individual points at 2 cm marked.

B) Wet Test

1. Soak the prepared test samples in distilled water at 23.0+/-1.0 ºC for 2 hours.
2. Remove the samples from the water, and test immediately following the directions outlined in “A) Dry test “above.

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CALCULATION:

Laminar Peel Strength (kgf/cm) = L / W

TEST REPORT:

1. Laminar strength - the **minimum** and **maximum** peaks to the nearest 0.1kgf/cm.
2. The mode of failure should be noted as follows:
   - **AF** = Failure at the adhesive /substrate interface of the adhesive/primer interface.
   - **CF** = Cohesive failure of the adhesive (failure within the adhesive or between the two layers of adhesive).
   - **PF** = Failure at the primer/substrate interface.
   - **SF** = Failure of the substrate, including the location of the failure within the substrate.
3. The reason for bond failure should also be reported, such as Poor Priming, Poor cementing, Over dried cement, Wet Cement, Wet primer, Poor pressing, Poor roughing, Weak material, Dusty Component, etc.